Yinyi Lin

yinyilin@hku.hk | Hong Kong | linyinyi

Summary

- Postdoctoral Fellow, Department of Geography, The University of Hong Kong. Geospatial scientist working on GeoAI and multisource remote sensing for urban environmental health, climate resilience, and sustainable urbanization. I integrate satellite observations, environmental big data, spatial statistics, and artificial intelligence to produce high-resolution, policy-relevant evidence across cities and regions worldwide.

Experience

The University of Hong Kong, Department of Geography, Postdoctoral Fellow — Hong Kong
2021 – present
5 years

United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Intern

Education

PhD **The Chinese University of Hong Kong**, Earth System and GeoInformation Science — Hong Kong
2021

BSc **Wuhan University**, Geographic Condition Monitoring — Wuhan, China
2017

Research Interests

GeoAI and multisource remote sensing: geospatial artificial intelligence, optical/SAR/LiDAR/hyperspectral/nighttime-light fusion, deep learning, Google Earth Engine

Urban environmental health: urban heat, air pollution, compound exposure, mortality, environmental inequality, ageing populations

Climate risk and community resilience: floods, droughts, extreme temperatures, compound risk, disaster-impact mapping, climate adaptation

Sustainable urbanization: urban expansion, land-cover change, urban-rural inequality, policy-relevant spatial evidence

Publications

Underestimated mortality risks from compound urban heat and air pollution islands in Global North cities
Yinyi Lin, et al.
doi.org/10.1038/s42949-026-00411-3

Incorporating synthetic aperture radar and optical images to investigate the annual dynamics of anthropogenic impervious surface at large scale
Yinyi Lin, et al.
doi.org/10.1016/j.rse.2020.111757

Leveraging optical and SAR data with a UU-Net for large-scale road extraction
Yinyi Lin, et al.
doi.org/10.1016/j.jag.2021.102498

A seasonal resilience index to evaluate the impacts of super typhoons on urban vegetation in Hong Kong
Yasong Guo, et al. (incl. Yinyi Lin)
doi.org/10.1080/24694452.2021.1989284

Coastal urbanization may indirectly positively impact growth of mangrove forests
Shan Wei, et al. (incl. Yinyi Lin)
doi.org/10.1038/s43247-024-01776-y

Projects

Compound urban heat, air pollution, and mortality in Global North cities (published)
Satellite-derived compound exposure assessment linked to mortality via structural equation modelling and multiscale analysis.

Multisource urbanization monitoring (published)
Long-term urban land-cover and expansion mapping from optical and SAR time series with machine and deep learning.

Teaching

GEOG7142: Big Data and GIS for China Development Studies: The University of Hong Kong — postgraduate course on geospatial big data, remote sensing, GIS workflows, Google Earth Engine, and spatial analysis. Taught in Spring 2023 and Spring 2024.

Professional Service

Youth Editorial Board Member: Chinese Geographical Science

Internal Coordinator: Hong Kong Geographical Association

Reviewer: international peer-reviewed journals in remote sensing, GIScience, and urban environment

Technical Expertise

Remote sensing & GeoAI: optical/SAR/LiDAR/hyperspectral/nighttime-light data, image classification and fusion, deep learning (CNNs, U-Net variants), machine learning (random forests, SVM)

Geospatial analysis: GIS, spatial statistics, structural equation modelling, time-series analysis, Google Earth Engine, cloud-scale processing